Biotechnology

Ever since the days of Rene Descartes, the French philosopher, mathematician and biologist of the seventeenth century, all human knowledge and especially the natural sciences were directed to develop technologies which add to the creature comforts of human lives, and value to human life.

How the approach changed

- the whole approach to understanding natural phenomena became anthropocentric
- physics and chemistry gave rise to engineering, technologies and industries which all worked for human comfort and welfare
- the major utility of the biological world is as a source of food

What biotechnology brought

- biotechnology, the twentieth century off-shoot of modern biology, changed our daily life
- its products brought qualitative improvement in health and food production
- the basic principles underlying biotechnological processes, and some applications, are highlighted and discussed in this unit

In this unit

Ch 9 Biotechnology: Principles and Processes

Ch 10 Biotechnology and Its Applications

Herbert Boyer, born 1936

- he was born in 1936 and brought up in a corner of western Pennsylvania where railroads and mines were the destiny of most young men
- he completed graduate work at the University of Pittsburgh in 1963, followed by three years of post-graduate studies at Yale
- in 1966 Boyer took over an assistant professorship at the University of California at San Francisco

The sticky ends

- by 1969 he performed studies on a couple of restriction enzymes of the E. coli bacterium with especially useful properties
- Boyer observed that these enzymes have the capability of cutting DNA strands in a particular fashion, which left what has become known as sticky ends on the strands
- these clipped ends made pasting together pieces of DNA a precise exercise

The conversation with Cohen

- this discovery led to a rich and rewarding conversation in Hawaii with a Stanford scientist named Stanley Cohen
- Cohen had been studying small ringlets of DNA called plasmids, which float about freely in the cytoplasm of certain bacterial cells and replicate independently from the coding strand of DNA, and had developed a method of removing these plasmids from the cell and then reinserting them in other cells
- combining this process with that of DNA splicing enabled Boyer and Cohen to recombine segments of DNA in desired configurations and insert the DNA in bacterial cells, which could then act as manufacturing plants for specific proteins
- this breakthrough was the basis upon which the discipline of biotechnology was founded

9.1 What biotechnology is

The broad and the narrow sense

- biotechnology deals with techniques of using live organisms, or enzymes from organisms, to produce products and processes useful to humans
- in this sense making curd, bread or wine, which are all microbe-mediated processes, could also be thought of as a form of biotechnology
- however the term is used in a restricted sense today, to refer to those processes which use genetically modified organisms to achieve the same on a larger scale, while many other processes and techniques are also included, for example in vitro fertilisation leading to a test-tube baby, synthesising a gene and using it, developing a DNA vaccine or correcting a defective gene

The EFB definition

- the European Federation of Biotechnology (EFB) has given a definition of biotechnology that encompasses both the traditional view and modern molecular biotechnology
- the definition given by EFB is the integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services

9.2 The two core techniques

Genetic engineering

- techniques to alter the chemistry of genetic material, that is DNA and RNA
- to introduce these into host organisms
- and thus change the phenotype of the host organism

Bioprocess engineering

- maintenance of a sterile, that is microbial contamination-free, ambience in chemical engineering processes
- to enable growth of only the desired microbe or eukaryotic cell in large quantities
- for the manufacture of biotechnological products such as antibiotics, vaccines and enzymes

Why genetic engineering was needed

- sexual reproduction provides opportunities for variation and the formulation of unique combinations of genetic setup, some of which may be beneficial to the organism as well as the population, so asexual reproduction preserves the genetic information while sexual reproduction permits variation
- traditional hybridisation procedures used in plant and animal breeding very often lead to inclusion and multiplication of undesirable genes along with the desired genes
- the techniques of genetic engineering, which include creation of recombinant DNA, use of gene cloning and gene transfer, overcome this limitation and allow us to isolate and introduce only one or a set of desirable genes without introducing undesirable ones into the target organism

9.3 The origin of replication

Why alien DNA needs it

- a piece of DNA somehow transferred into an alien organism would most likely not be able to multiply itself in the progeny cells, but when it gets integrated into the genome of the recipient it may multiply and be inherited along with the host DNA, because the alien piece has become part of a chromosome, which has the ability to replicate
- in a chromosome there is a specific DNA sequence called the origin of replication, which is responsible for initiating replication, so for the multiplication of any alien piece of DNA in an organism it needs to be part of a chromosome which has that sequence
- thus an alien DNA is linked with the origin of replication, so that it can replicate and multiply itself in the host organism, which can also be called cloning, or making multiple identical copies of any template DNA

The first recombinant DNA

- the construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid, that is an autonomously replicating circular extra-chromosomal DNA, of Salmonella typhimurium
- Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene, cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance
- the cutting of DNA at specific locations became possible with the discovery of the so-called molecular scissors, the restriction enzymes

Vectors and ligase

- the cut piece of DNA was then linked with the plasmid DNA, and these plasmid DNA act as vectors to transfer the piece of DNA attached to them, just as a mosquito acts as an insect vector to transfer the malarial parasite into the human body
- the linking of the antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase, which acts on cut DNA molecules and joins their ends, making a new combination of circular autonomously replicating DNA created in vitro, known as recombinant DNA
- when this DNA is transferred into Escherichia coli, a bacterium closely related to Salmonella, it could replicate using the new host's DNA polymerase enzyme and make multiple copies, and this ability to multiply copies of the antibiotic resistance gene in E. coli was called cloning of that gene in E. coli

Three basic steps

- identification of DNA with desirable genes
- introduction of the identified DNA into the host
- maintenance of introduced DNA in the host and transfer of the DNA to its progeny

9.4 Restriction enzymes

Hind II, and how they are named

- genetic engineering can be accomplished only if we have the key tools, that is restriction enzymes, polymerase enzymes, ligases, vectors and the host organism
- in 1963 the two enzymes responsible for restricting the growth of bacteriophage in Escherichia coli were isolated, one of these adding methyl groups to DNA while the other cut DNA, the latter being called restriction endonuclease
- the first restriction endonuclease, Hind II, whose functioning depended on a specific DNA nucleotide sequence, was isolated and characterised five years later, and it was found that it always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs, known as the recognition sequence for Hind II, while today we know more than 900 restriction enzymes isolated from over 230 strains of bacteria, each recognising different recognition sequences
- the convention for naming these enzymes is that the first letter comes from the genus and the second two letters from the species of the prokaryotic cell from which they were isolated, so EcoRI comes from Escherichia coli RY 13, the letter R being derived from the name of the strain

EcoRI inspecting the length of the DNA

Its recognition sequence found, the enzyme binds

Both strands cut in their sugar-phosphate backbones

Single stranded overhangs left, the sticky ends

Figure 9.A Steps in formation of recombinant DNA by action of restriction endonuclease enzyme - EcoRI

9.5 Nucleases and palindromes

Two kinds of nuclease

- roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria
- restriction enzymes belong to a larger class of enzymes called nucleases, which are of two kinds, exonucleases and endonucleases
- exonucleases remove nucleotides from the ends of the DNA, whereas endonucleases make cuts at specific positions within the DNA

How a restriction endonuclease works

- each restriction endonuclease functions by inspecting the length of a DNA sequence
- once it finds its specific recognition sequence it will bind to the DNA and cut each of the two strands of the double helix at specific points in their sugar-phosphate backbones
- each restriction endonuclease recognises a specific palindromic nucleotide sequence in the DNA

What a palindrome is

- palindromes are groups of letters that form the same words when read both forward and backward, for example MALAYALAM
- as against a word-palindrome, where the same word is read in both directions, the palindrome in DNA is a sequence of base pairs that reads the same on the two strands when the orientation of reading is kept the same

Source DNA and vector DNA cut with the same enzyme

Both carrying the same kind of sticky ends

The two joined end to end by DNA ligase

A recombinant DNA molecule, taken up by the cloning host

Figure 9.B Diagrammatic representation of recombinant DNA technology

9.6 Sticky ends

The example sequence

- the sequence 5' GAATTC 3' on one strand, against 3' CTTAAG 5' on the other, reads the same on the two strands in the 5' to 3' direction, and this is also true if read in the 3' to 5' direction

Why the ends are sticky

- restriction enzymes cut the strand of DNA a little away from the centre of the palindrome sites, but between the same two bases on the opposite strands, which leaves single stranded portions at the ends
- there are overhanging stretches called sticky ends on each strand, named so because they form hydrogen bonds with their complementary cut counterparts
- this stickiness of the ends facilitates the action of the enzyme DNA ligase

Making a recombinant molecule

- restriction endonucleases are used in genetic engineering to form recombinant molecules of DNA, which are composed of DNA from different sources or genomes
- when cut by the same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends and can be joined together end-to-end using DNA ligases

Undigested DNA in the first lane

Digested sets of DNA fragments in the later lanes

Smaller fragments moving farther towards the anode

Bands seen orange under UV after staining

Figure 9.C A typical agarose gel electrophoresis showing migration of undigested (lane 1) and digested set of DNA fragments (lane 2 to 4)

9.7 Separation and isolation of DNA fragments

Gel electrophoresis

- unless one cuts the vector and the source DNA with the same restriction enzyme, the recombinant vector molecule cannot be created
- the cutting of DNA by restriction endonucleases results in fragments of DNA, which can be separated by a technique known as gel electrophoresis, since DNA fragments are negatively charged molecules and can be separated by forcing them to move towards the anode under an electric field through a medium or matrix
- nowadays the most commonly used matrix is agarose, a natural polymer extracted from sea weeds, and the DNA fragments separate according to their size through the sieving effect provided by the gel, so the smaller the fragment size the farther it moves
- the separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide followed by exposure to UV radiation, when bright orange coloured bands of DNA are seen, and the separated bands are cut out from the agarose gel and extracted from the gel piece, a step known as elution, the purified fragments then being used in constructing recombinant DNA by joining them with cloning vectors

9.8 Cloning vectors

Why plasmids and phages

- plasmids and bacteriophages have the ability to replicate within bacterial cells independent of the control of chromosomal DNA
- bacteriophages, because of their high number per cell, have very high copy numbers of their genome within the bacterial cells, while some plasmids may have only one or two copies per cell whereas others may have 15-100 copies per cell, and their numbers can go even higher
- if we are able to link an alien piece of DNA with bacteriophage or plasmid DNA we can multiply its numbers equal to the copy number of the plasmid or bacteriophage, and vectors used at present are engineered in such a way that they help easy linking of foreign DNA and selection of recombinants from non-recombinants

pBR322, an E. coli cloning vector

Restriction sites for Hind III, EcoR I, BamH I, Sal I, Pvu II, Pst I and Cla I

ori, the origin of replication

ampR and tetR, the two antibiotic resistance genes

rop, coding for proteins involved in replication

Figure 9.D E. coli cloning vector pBR322 showing restriction sites (Hind III, EcoR I, BamH I, Sal I, Pvu II, Pst I, Cla I), ori and antibiotic resistance genes (ampR and tetR). rop codes for the proteins involved in the replication of the plasmid.

9.9 What a cloning vector needs

Origin of replication and selectable marker

- origin of replication is a sequence from where replication starts, and any piece of DNA linked to it can be made to replicate within the host cells, while this sequence is also responsible for controlling the copy number of the linked DNA, so if one wants to recover many copies of the target DNA it should be cloned in a vector whose origin supports a high copy number
- in addition, the vector requires a selectable marker, which helps in identifying and eliminating non-transformants and selectively permitting the growth of the transformants, transformation being a procedure through which a piece of DNA is introduced in a host bacterium
- normally the genes encoding resistance to antibiotics such as ampicillin, chloramphenicol, tetracycline or kanamycin are considered useful selectable markers for E. coli, since the normal E. coli cells do not carry resistance against any of these antibiotics

Cloning sites

- in order to link the alien DNA the vector needs to have very few, preferably single, recognition sites for the commonly used restriction enzymes, since the presence of more than one recognition site within the vector will generate several fragments, which will complicate the gene cloning
- the ligation of alien DNA is carried out at a restriction site present in one of the two antibiotic resistance genes, so one can ligate a foreign DNA at the BamH I site of the tetracycline resistance gene in the vector pBR322
- the recombinant plasmids will lose tetracycline resistance due to insertion of foreign DNA but can still be selected out from non-recombinant ones by plating the transformants on ampicillin containing medium, the transformants growing there then being transferred to a medium containing tetracycline
- the recombinants will grow in ampicillin containing medium but not on that containing tetracycline, while non-recombinants will grow on the medium containing both antibiotics, so one antibiotic resistance gene helps in selecting the transformants whereas the other gets inactivated due to insertion of alien DNA and helps in selection of recombinants

9.10 Selecting recombinants by colour

Insertional inactivation

- selection of recombinants due to inactivation of antibiotics is a cumbersome procedure, because it requires simultaneous plating on two plates having different antibiotics
- therefore alternative selectable markers have been developed which differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate
- in this, a recombinant DNA is inserted within the coding sequence of an enzyme, β-galactosidase, which results in inactivation of the gene for synthesis of this enzyme, referred to as insertional inactivation
- the presence of a chromogenic substrate gives blue coloured colonies if the plasmid in the bacteria does not have an insert, while the presence of an insert results in insertional inactivation of the β-galactosidase gene and the colonies do not produce any colour, so these are identified as recombinant colonies

9.11 Vectors for plants and animals

Learning from pathogens

- we have learnt the lesson of transferring genes into plants and animals from bacteria and viruses, which have known for ages how to deliver genes to transform eukaryotic cells and force them to do what the bacteria or viruses want
- for example Agrobacterium tumifaciens, a pathogen of several dicot plants, is able to deliver a piece of DNA known as T-DNA to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen, while similarly retroviruses in animals have the ability to transform normal cells into cancerous cells
- a better understanding of the art of delivering genes by pathogens in their eukaryotic hosts has generated knowledge to transform these tools of pathogens into useful vectors for delivering genes of interest to humans
- the tumor inducing (Ti) plasmid of Agrobacterium tumifaciens has now been modified into a cloning vector which is no more pathogenic to the plants but is still able to use the mechanisms to deliver genes of our interest into a variety of plants, and similarly retroviruses have been disarmed and are now used to deliver desirable genes into animal cells

9.12 Competent host

Making the bacteria competent

- since DNA is a hydrophilic molecule it cannot pass through cell membranes
- in order to force bacteria to take up the plasmid, the bacterial cells must first be made competent to take up DNA, which is done by treating them with a specific concentration of a divalent cation such as calcium, which increases the efficiency with which DNA enters the bacterium through pores in its cell wall

Cells broken open to release the DNA

RNA removed by ribonuclease, protein by protease

Purified DNA precipitated with chilled ethanol

Fine threads collected by spooling

Figure 9.E DNA that separates out can be removed by spooling

9.13 Four ways in

Heat shock, and the alternatives

- recombinant DNA can be forced into competent cells by incubating the cells with recombinant DNA on ice, followed by placing them briefly at 42 degrees C, that is a heat shock, and then putting them back on ice, which enables the bacteria to take up the recombinant DNA
- in a method known as micro-injection, recombinant DNA is directly injected into the nucleus of an animal cell
- in another method suitable for plants, cells are bombarded with high velocity micro-particles of gold or tungsten coated with DNA, a method known as biolistics or gene gun
- the last method uses disarmed pathogen vectors, which when allowed to infect the cell transfer the recombinant DNA into the host

9.14 Isolation of the genetic material

Getting pure DNA

- recombinant DNA technology involves several steps in a specific sequence, isolation of DNA, fragmentation of DNA by restriction endonucleases, isolation of a desired DNA fragment, ligation of the fragment into a vector, transferring the recombinant DNA into the host, culturing the host cells in a medium at large scale, and extraction of the desired product
- in order to cut the DNA with restriction enzymes it needs to be in pure form, free from other macro-molecules, and since the DNA is enclosed within membranes we have to break the cell open to release DNA along with other macromolecules such as RNA, proteins, polysaccharides and lipids, which can be achieved by treating the bacterial cells or plant or animal tissue with enzymes such as lysozyme for bacteria, cellulase for plant cells and chitinase for fungus
- genes are located on long molecules of DNA interwined with proteins such as histones, and the RNA can be removed by treatment with ribonuclease whereas proteins can be removed by treatment with protease, other molecules being removed by appropriate treatments, and purified DNA ultimately precipitates out after the addition of chilled ethanol, which can be seen as a collection of fine threads in the suspension

Denaturation, the double strand separated by heat

Primer annealing, two primers binding the single strands

Extension of primers by DNA polymerase

The cycle repeated, the segment amplified

Figure 9.F Polymerase chain reaction (PCR) : Each cycle has three steps: (i) Denaturation; (ii) Primer annealing; and (iii) Extension of primers

9.15 Cutting DNA at specific locations

Digestion and ligation

- restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzyme at the optimal conditions for that specific enzyme, and agarose gel electrophoresis is employed to check the progression of a digestion, DNA being a negatively charged molecule and hence moving towards the positive electrode
- the process is repeated with the vector DNA also
- after having cut the source DNA as well as the vector DNA with a specific restriction enzyme, the cut out gene of interest from the source DNA and the cut vector with space are mixed and ligase is added, which results in the preparation of recombinant DNA

9.16 Amplification by PCR

What PCR does

- PCR stands for polymerase chain reaction, in which multiple copies of the gene or DNA of interest are synthesised in vitro using two sets of primers, small chemically synthesised oligonucleotides that are complementary to the regions of DNA, and the enzyme DNA polymerase
- the enzyme extends the primers using the nucleotides provided in the reaction and the genomic DNA as template

9.17 Amplification, and the transformed cell

A billion copies

- if the process of replication of DNA is repeated many times the segment of DNA can be amplified to approximately a billion times, that is 1 billion copies are made
- such repeated amplification is achieved by the use of a thermostable DNA polymerase, isolated from a bacterium, Thermus aquaticus, which remains active during the high temperature induced denaturation of double stranded DNA
- the amplified fragment, if desired, can now be used to ligate with a vector for further cloning

Selecting the transformants

- there are several methods of introducing the ligated DNA into recipient cells, and recipient cells, after being made competent to receive, take up DNA present in their surroundings
- so if a recombinant DNA bearing a gene for resistance to an antibiotic such as ampicillin is transferred into E. coli cells, the host cells become transformed into ampicillin-resistant cells
- if we spread the transformed cells on agar plates containing ampicillin, only transformants will grow and untransformed recipient cells will die, so the ampicillin resistance gene in this case is called a selectable marker

9.18 Obtaining the foreign gene product

Expression, and scale

- when a piece of alien DNA is inserted into a cloning vector and transferred into a bacterial, plant or animal cell, the alien DNA gets multiplied, but in almost all recombinant technologies the ultimate aim is to produce a desirable protein, so there is a need for the recombinant DNA to be expressed, the foreign gene getting expressed under appropriate conditions
- the expression of foreign genes in host cells involves understanding many technical details
- after having cloned the gene of interest and optimised the conditions to induce the expression of the target protein, one has to consider producing it on a large scale, and if any protein encoding gene is expressed in a heterologous host it is called a recombinant protein
- the cells harbouring cloned genes of interest may be grown on a small scale in the laboratory, and the cultures may be used for extracting the desired protein and then purifying it by using different separation techniques

A simple stirred-tank bioreactor

A sparged stirred-tank bioreactor, air bubbled through

An agitator system, and an oxygen delivery system

Foam, temperature and pH control systems

Sampling ports for withdrawing small volumes

Figure 9.G (a) Simple stirred-tank bioreactor; (b) Sparged stirred-tank bioreactor through which sterile air bubbles are sparged

9.19 Growing the cells at scale

Continuous culture

- the cells can also be multiplied in a continuous culture system, wherein the used medium is drained out from one side while fresh medium is added from the other, to maintain the cells in their physiologically most active log or exponential phase
- this type of culturing method produces a larger biomass, leading to higher yields of the desired protein
- small volume cultures cannot yield appreciable quantities of products, so to produce in large quantities the development of bioreactors, where large volumes of 100-1000 litres of culture can be processed, was required

What a bioreactor is

- bioreactors can be thought of as vessels in which raw materials are biologically converted into specific products, individual enzymes and the like, using microbial, plant, animal or human cells
- a bioreactor provides the optimal conditions for achieving the desired product by providing optimum growth conditions of temperature, pH, substrate, salts, vitamins and oxygen
- the most commonly used bioreactors are of the stirring type, and a stirred-tank reactor is usually cylindrical or with a curved base to facilitate the mixing of the reactor contents, the stirrer facilitating even mixing and oxygen availability throughout, while alternatively air can be bubbled through the reactor
- the bioreactor has an agitator system, an oxygen delivery system, a foam control system, a temperature control system, a pH control system and sampling ports, so that small volumes of the culture can be withdrawn periodically

9.20 Downstream processing

After the biosynthetic stage

- the product has to be subjected to a series of processes before it is ready for marketing as a finished product, and these processes include separation and purification, collectively referred to as downstream processing
- the product has to be formulated with suitable preservatives, and such formulation has to undergo thorough clinical trials as in the case of drugs
- strict quality control testing for each product is also required, and the downstream processing and quality control testing vary from product to product

9.21 Summary

What the chapter has covered

- biotechnology deals with large scale production and marketing of products and processes using live organisms, cells or enzymes
- modern biotechnology using genetically modified organisms was made possible only when man learnt to alter the chemistry of DNA and construct recombinant DNA, a key process called recombinant DNA technology or genetic engineering
- this process involves the use of restriction endonucleases, DNA ligase and appropriate plasmid or viral vectors to isolate and ferry the foreign DNA into host organisms, expression of the foreign gene, purification of the gene product, that is the functional protein, and finally making a suitable formulation for marketing
- large scale production involves the use of bioreactors

9.22 Exercises

From this chapter

1. Can you list 10 recombinant proteins which are used in medical practice? Find out where they are used as therapeutics (use the internet).
2. Make a chart (with diagrammatic representation) showing a restriction enzyme, the substrate DNA on which it acts, the site at which it cuts DNA and the product it produces.
3. From what you have learnt, can you tell whether enzymes are bigger or DNA is bigger in molecular size? How did you know?
4. What would be the molar concentration of human DNA in a human cell? Consult your teacher.
5. Do eukaryotic cells have restriction endonucleases? Justify your answer.
6. Besides better aeration and mixing properties, what other advantages do stirred tank bioreactors have over shake flasks?
7. Collect 5 examples of palindromic DNA sequences by consulting your teacher. Better try to create a palindromic sequence by following base-pair rules.
8. Can you recall meiosis and indicate at what stage a recombinant DNA is made?
9. Can you think and answer how a reporter enzyme can be used to monitor transformation of host cells by foreign DNA in addition to a selectable marker?

9.23 Exercises, continued

From this chapter, continued

1. Describe briefly the following: (a) Origin of replication (b) Bioreactors (c) Downstream processing
2. Explain briefly (a) PCR (b) Restriction enzymes and DNA (c) Chitinase
3. Discuss with your teacher and find out how to distinguish between (a) Plasmid DNA and Chromosomal DNA (b) RNA and DNA (c) Exonuclease and Endonuclease